Randomized Linear Algebra

Approximating Matrix Multiplication

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Matrix Multiplication is Expensive

If A is $m \times n$ and B is $n \times p$, calculating AB is expensive.

$$\underbrace{\begin{bmatrix} \cdot & \cdots & \cdot \\ \vdots & \ddots & \vdots \\ \cdot & \cdots & \cdot \end{bmatrix}}_{\substack{m \times n \\ A}} \times \underbrace{\begin{bmatrix} \cdot & \cdots & \cdot \\ \vdots & \ddots & \vdots \\ \cdot & \cdots & \cdot \end{bmatrix}}_{\substack{n \times p \\ B}} = \underbrace{\begin{bmatrix} \cdot & \cdots & \cdot \\ \vdots & \ddots & \vdots \\ \cdot & \cdots & \cdot \end{bmatrix}}_{\substack{m \times p \\ AB}}$$

Total: mnp multiplications.

AB as a Sum of Outer Products

Recall that AB can be written as the sum of rank-1 matrices.

$$AB = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix} \begin{bmatrix} - \, b_1^\top \, - \\ \vdots \\ - \, b_n^\top \, - \end{bmatrix}$$

$$= \underbrace{\underbrace{\begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot \end{bmatrix}}_{m \times p} + \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot \end{bmatrix} + \cdots + \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot \end{bmatrix}}_{n \text{ total}}$$

Sampling a Few Rank-1 Pieces of AB

We can estimate AB by sampling a few of these rank-1 matrices.

$$AB = \underbrace{\begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot \end{bmatrix} + {\color{red}\begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot \end{bmatrix}} + \cdots + \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot \end{bmatrix}}_{n \text{ total, } {\color{red} s \text{ samples}}}$$

This reduces the number of computations by a factor of s/n!

Estimation for AB with Uniform Sampling

Since we had s samples, we can estimate AB as:

$$AB = \frac{n}{s} \sum_{j=1}^{s} a_j b_j^{\top}.$$

Issue

Imagine we had a few relatively large $a_j b_j^{\top}$ that contribute a lot to AB.
They should receive more weight compared to the other outer products!

Estimation for AB with Uniform Sampling

Since we had s samples, we can estimate AB as:

$$AB = \frac{n}{s} \sum_{j=1}^{s} a_j b_j^{\mathsf{T}}.$$

Issue

Imagine we had a few relatively large $a_j b_j^{\mathsf{T}}$ that contribute a lot to AB.
They should receive more weight compared to the other outer products!

Non-Uniform Sampling

- Instead of a uniform distribution, we will give each rank-1 matrix a weight.
- Assign probabilities p_k to each $a_k b_k^T$ with $p_1 + \cdots + p_n = 1$ and $k = 1, 2, \cdots, n$.
- Sample s rank-1 matrices, with replacement.
- Multiply each of a_k and b_k^T by $\frac{1}{\sqrt{sp_k}}$ to normalize.
- For each k in the s chosen, we will add $\frac{a_k}{\sqrt{sp_k}} \cdot \frac{b_k^T}{\sqrt{sp_k}} = \frac{a_k b_k^T}{sp_k}$ into the estimation of AB.

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Sample Expected Value

For each of the s samples, we pick $\frac{a_k b_k^\top}{s p_k}$ with probability p_k, with $k = 1, 2, \cdots, n$.

The expected value for each sample X is thus:

$$\begin{aligned} \mathbb{E}[X] &= \sum_{k=1}^{n} p_k \cdot \frac{a_k b_k^\top}{s p_k} \\ &= \frac{1}{s} \sum_{k=1}^{n} a_k b_k^\top. \end{aligned}$$

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Sum of Samples Expected Value

Estimation for given p_k

Thus, for any weights p_k we choose, the expected value of the sum of all s samples is

$$s \cdot \mathbb{E}[X] = \sum_{k=1}^{n} a_k b_k^\top = AB.$$

We always get the same expected value of AB!

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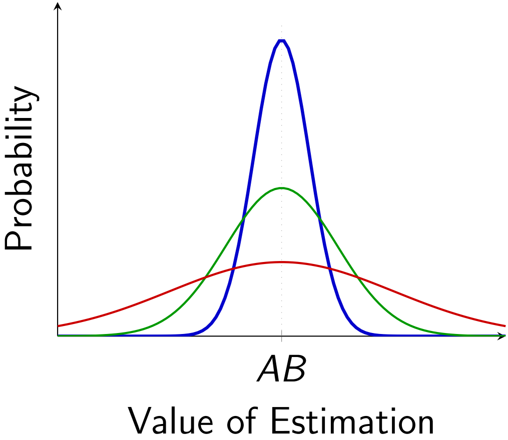
Visualizing Variance

Various choices of p_k will give different variances.
How can we pick the best p_k?



Minimizing Variance

- We would like to find the p_k to minimize the variance.
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Calculating Variance: Big Picture

- Let's pick s samples, each of the form $\frac{a_t b_t^\mathsf{T}}{sp_t}$.
- Select an entry ij to focus on over all the samples.
- Define random variable X_t to be the ij entry of the t-th sample.
- Calculate the variance of X_t.
- Then, the ij term of estimated matrix will have variance of the sum of the variances on X.
- Finally, the variance of the whole estimated matrix is the sum of the variances of all entries in the matrix.

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Variance for General p_k

Fix i, j. Define X_t for $t = 1, 2, \cdots, s$ to be $\left(\frac{a_t b_t^\mathsf{T}}{sp_t}\right)_{ij}$.

Each entry's variance from AB_{ij} is by definition $\mathrm{Var}\left[X_t\right] = \mathbb{E}[X_t^2] - (\mathbb{E}[X_t])^2$.

$$\mathrm{Var}\left[X_t\right] = \sum_{k=1}^{n} p_k \cdot \left(\frac{a_k b_k^\mathsf{T}}{sp_k}\right)_{ij}^2 - \left(\sum_{k=1}^{n} p_k \cdot \left(\frac{a_k b_k^\mathsf{T}}{sp_k}\right)_{ij}\right)^2.$$

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Variance for General p_k

We first simplify $(\mathbb{E}[X_t])^2$ by noting the sum of rank-1 matrices.

$$\begin{aligned}
\operatorname{Var}[X_t] &= \sum_{k=1}^{n} p_k \cdot \left(\frac{a_k b_k^\top}{s p_k}\right)_{ij}^2 - \frac{1}{s^2}\left(\sum_{k=1}^{n} \left(a_k b_k^\top\right)_{ij}\right)^2 \\
&= \sum_{k=1}^{n} p_k \cdot \left(\frac{a_k b_k^\top}{s p_k}\right)_{ij}^2 - \frac{1}{s^2}\left(AB_{ij}\right)^2
\end{aligned}$$

Variance for General p_k

Note that $\left(a_k b_k^{\mathsf{T}}\right)_{ij}$ is $a_{ik} b_{kj}^{\mathsf{T}}$:

$$\begin{bmatrix} \cdot \\ \cdot \\ \cdot \\ \boxed{a_{ik}} \\ \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot & \boxed{b_{kj}^{\mathsf{T}}} & \cdot \end{bmatrix} \implies \left(\frac{a_k b_k^{\mathsf{T}}}{s p_k}\right)_{ij}^2 = \frac{(a_{ik})^2 (b_{kj}^{\mathsf{T}})^2}{s^2 p_k^2}.$$

Variance of X_t for fixed i, j

$\mathrm{Var}\left[X_t\right] = \frac{1}{s^2} \sum_{k=1}^{n} \frac{(a_{ik})^2 (b_{kj}^{\mathsf{T}})^2}{p_k} - \frac{1}{s^2} \left(AB_{ij}\right)^2$

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Variance for General p_k

Let Z be the estimated matrix. Then,

Variance of the ij term of Z

$$\begin{aligned}
\operatorname{Var}\left[Z_{ij}\right] &= \sum_{t=1}^{s} \operatorname{Var}\left[X_t\right] \\
&= s \operatorname{Var}\left[X_t\right] \\
&= \frac{1}{s} \sum_{k=1}^{n} \frac{(a_{ik})^2 (b_{kj}^{\mathsf{T}})^2}{p_k} - \frac{1}{s} \left(AB_{ij}\right)^2
\end{aligned}$$

Variance for General p_k

We see that the variance on Z will be the sum of the total variance of all the matrix entries.

- It involves the sum of the squares over all entries in AB.
- It involves the pairwise products of squared entries of vectors a_k and $b_k^\top$.

Variance for General p_k

Let us define some useful notation for simplicity.

Definition

The Frobenius norm is defined for matrix A as

$$||A||_F^2 = \sum_i \sum_j (A_{ij})^2.$$

Definition

For vector $\mathbf{x}$ with components $x_1, x_2, \cdots, x_n$, the square of its ℓ_2 norm is defined as:

$$||\mathbf{x}||_2^2 = x_1^2 + x_2^2 + \cdots + x_n^2.$$

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Variance for General p_k

Thus, the variance on Z for general p_k is:

Variance of Z

$$\sum_i \sum_j \mathrm{Var}\left[Z_{ij}\right] = \frac{1}{s}\left(\sum_{k=1}^{n} \frac{\|a_k\|_2^2 \, \|b_k^\mathsf{T}\|_2^2}{p_k} - \|AB\|_F^2\right).$$

Now, we can figure out how to minimize this variance!

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Minimizing Variance

Constraint of $\sum_{k=1}^{n} p_k = 1$ suggests Lagrange Multipliers!

For some constant λ,

$$L = \frac{1}{s}\left(\sum_{k=1}^{n} \frac{\|a_k\|_2^2 \, \|b_k^{\mathsf{T}}\|_2^2}{p_k} - \|AB\|_F^2\right) + \lambda\left(\sum_{k=1}^{n} p_k - 1\right).$$

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Minimizing Variance

Taking partial derivatives $\frac{\partial L}{\partial p_k}$ and setting to zero,

$$\frac{\|a_k\|_2^2 \, \|b_k^\mathsf{T}\|_2^2}{s p_k^2} = \lambda \implies p_k = \frac{\|a_k\|_2 \, \|b_k^\mathsf{T}\|_2}{\sqrt{s\lambda}}.$$

Minimizing Variance

Since $\sum_{k=1}^{n} p_k = 1$, we must have

$$\lambda = \frac{(\sum_{k=1}^{n} ||a_k||_2 \cdot ||b_k^{\mathsf{T}}||_2)^2}{s}.$$

Norm-squared Sampling

Thus, the weight on the kth rank-1 matrix is:

$$p_k = \frac{||a_k||_2 \cdot ||b_k^{\mathsf{T}}||_2}{\sum_{k=1}^{n} ||a_k||_2 \cdot ||b_k^{\mathsf{T}}||_2}.$$

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What's the minimum?

Definition

For simplicity, define C so that:

$$p_j = \frac{||a_j||_2 \cdot ||b_j^{\mathsf{T}}||_2}{C}, \;\; C = \sum_{j=1}^{n} ||a_j||_2 \cdot ||b_j^{\mathsf{T}}||_2.$$

Variance of Z

Summing $\mathrm{Var}[Z_{ij}]$ over i and j and plugging in p_k,

$$\sum_i \sum_j \mathrm{Var}\left[Z_{ij}\right] = \frac{1}{s} \cdot C^2 - \frac{1}{s}||AB||_F^2.$$

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Summary

- For very large matrices A and B, matrix multiplication is expensive.
- We solve this problem by sampling rank-1 matrices.
- Weighting the probabilities of selecting each rank-1 matrix does not change the expected value AB.
- We can minimize how spread out our answers are with norm-squared variance, which is the intuitive solution!

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